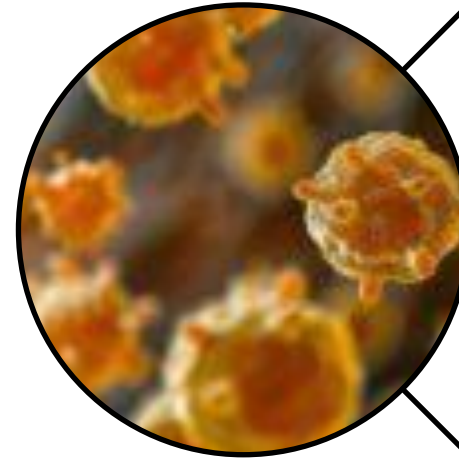


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**Antimicrobial
Resistance of
Bacterial Pathogens –
Africa**

8-13 March 2026

NICD, Johannesburg,
South Africa

Augusto Messa Jr.

CP1 – Introduction to Computational Methods

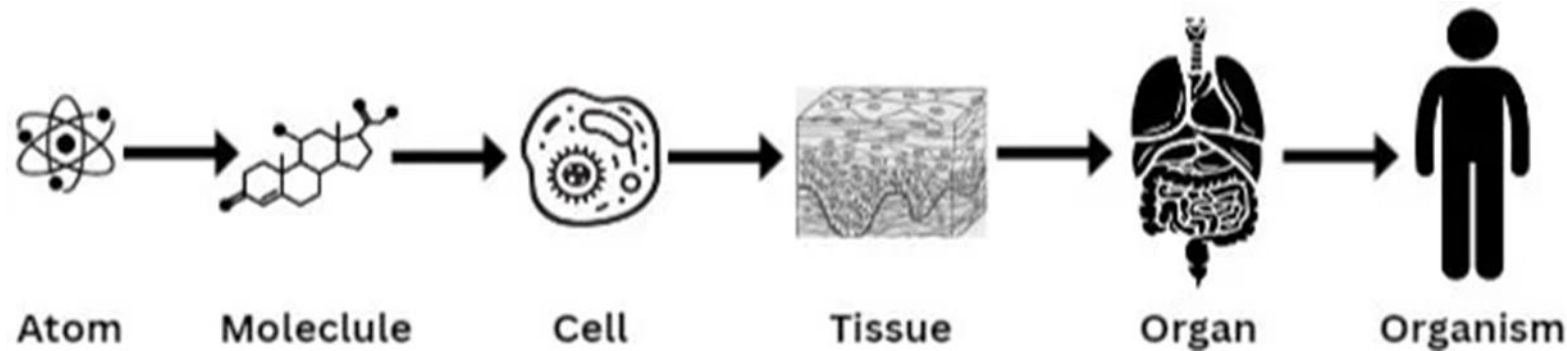
March 2026

Learning objectives

- Provide a brief overview of Computational Methods for Biological Applications.
- Provide a brief overview of the workflow from sequences to AMR report.
- Explain some essential data formats: FASTA, FASTQ, SAM/BAM, VCF/BCF, GFF.
- Disclaimer: This is not (nor pretends to be) a comprehensive introduction to Computational Biology or Computational Methods for Biology, etc.

Computational biology

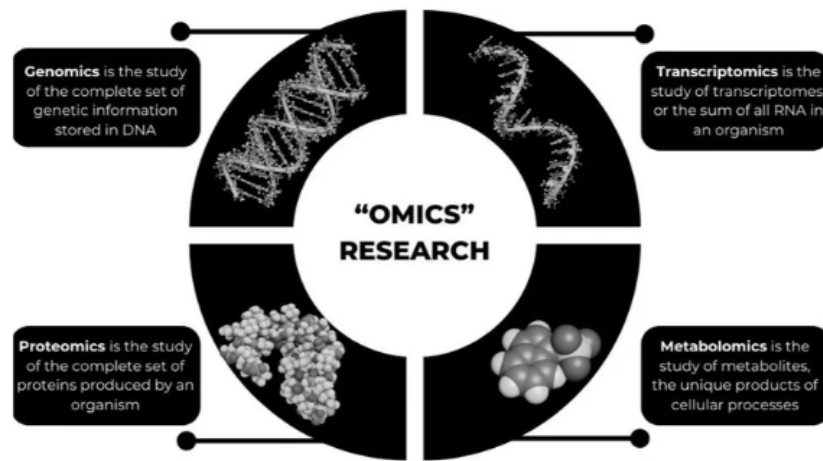
- Computational biology merges **computer science, mathematics (and statistics), and biology** to tackle complex biological puzzles.
 - **Using computers to analyze complex biological data** to gain insights on the biology of things: DNA sequences, protein structures.



DNA, RNAs, proteins, lipids, carbohydrates, etc.

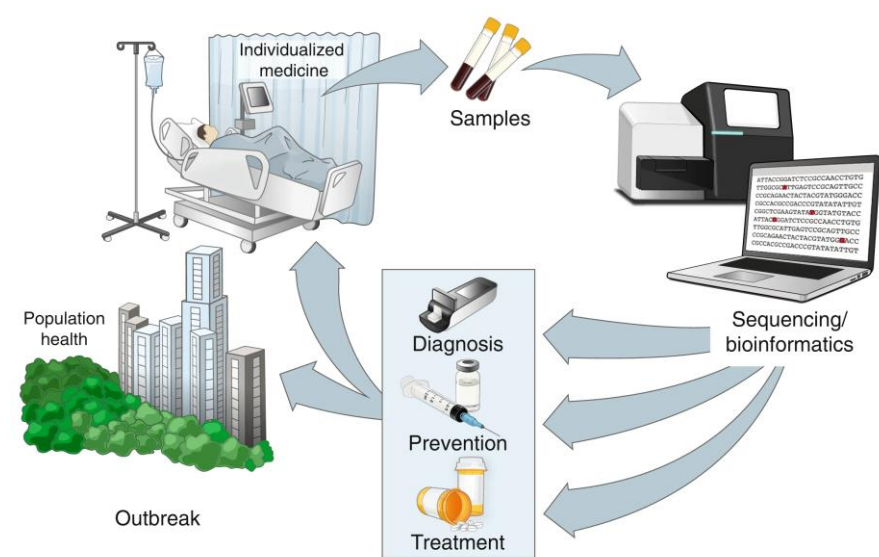
Computational biology - caveats

- **CB does not (and cannot) replace traditional biological lab research;** it complements it by helping translate large data into a testable hypothesis in labs.
- **Golden rule of computational biology: garbage in, garbage out.**
- **Our results are only as good as our hypotheses, the quality of samples, and the quality of data.**



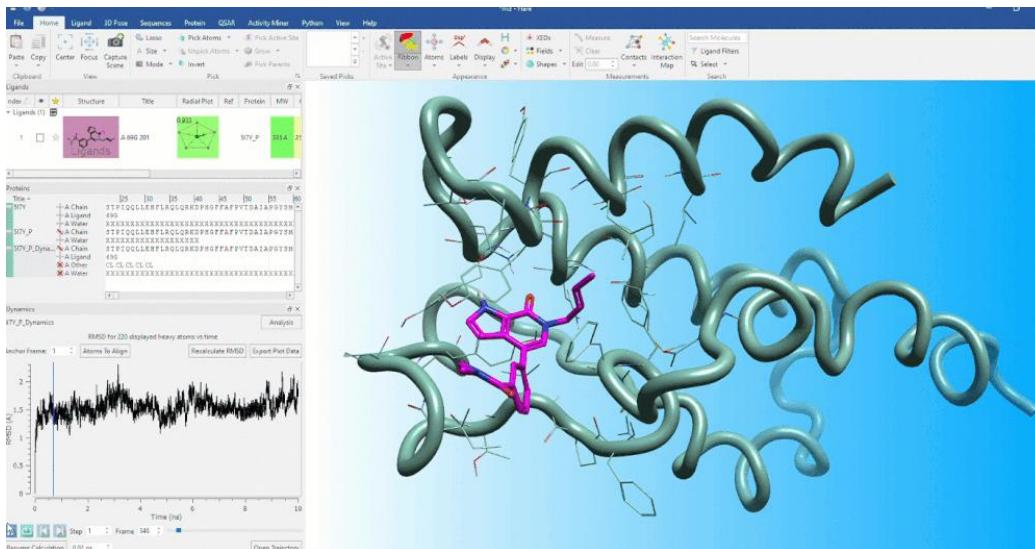
"Omics" Research

Simplify the analysis of large-scale biological data sets, which is generated by the growing interest in "omics" (genomics, transcriptomics, proteomics, metabolomics, etc.)



Public Health

Track pathogen evolution during outbreaks by analyzing sequenced samples to identify pathogens or variants and enable public health officials determine the most effective course of action. Ladner et al. 2019 (doi: 10.1038/s41591-019-0345-2)



Molecular dynamics simulations

Provide detailed information about molecular conformations, interactions, and dynamics. Can be used for drug discovery, etc.

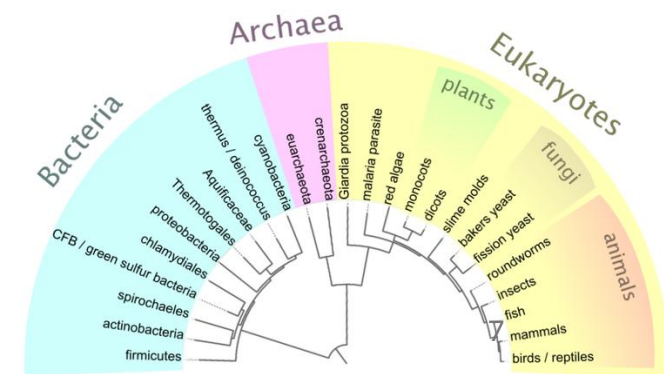
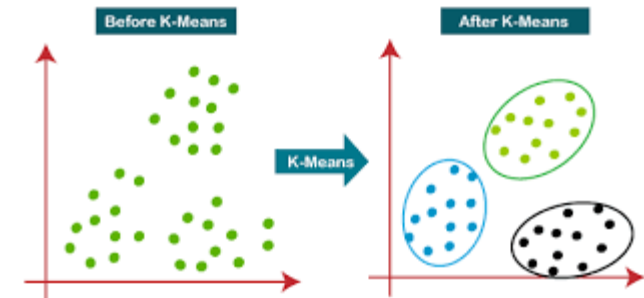
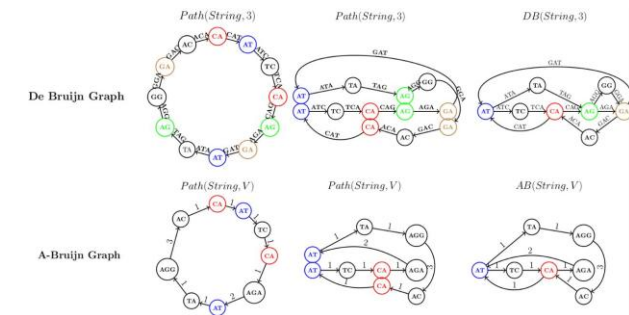
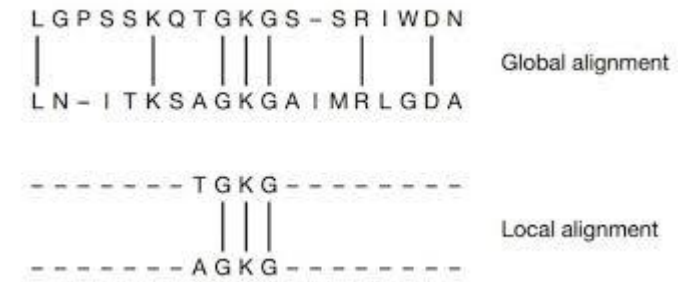


Pharmacogenomics

Provide insights into how certain genetic variants impact an individual's response to certain medications to help develop personalized treatment options.

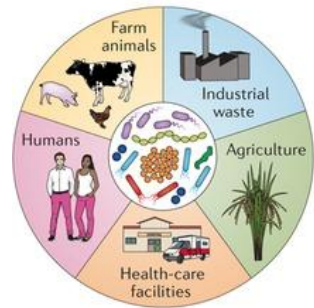
Basic algorithms, tools and techniques in CB

- **Sequence alignment algorithms:** arrange and compare sequences and identify regions of similarity (functional, structural or evolutionary relationships). E.g Needleman-Wunsch, Smith-Waterman, **BLAST**.
- **Sequence assembly algorithms:** reconstruct the complete sequence from short fragments (reads). E.g. de Bruijn graph-based methods.
- **Clustering algorithms:** group similar biological entities based on properties or measurements. E.g. hierarchical clustering, k-means.
- **ML algorithms:** build predictive models and identify patterns.
- **Phylogenetic inference algorithms:** reconstruct evolutionary relationships among organisms or sequences. E.g. Neighbor-Joining, Maximum-likelihood, Bayesian methods.

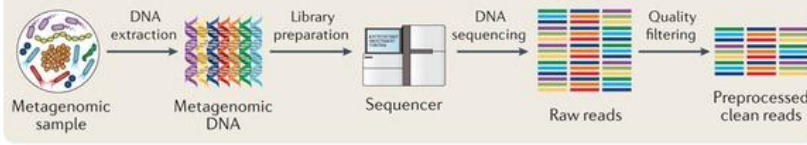


Assembly- vs. read-mapping-based approaches to study AMR using genomic technologies

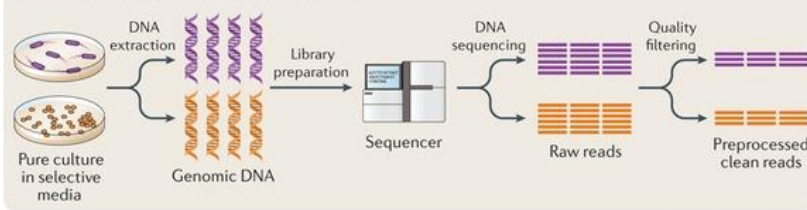
a Sample collection and sequencing



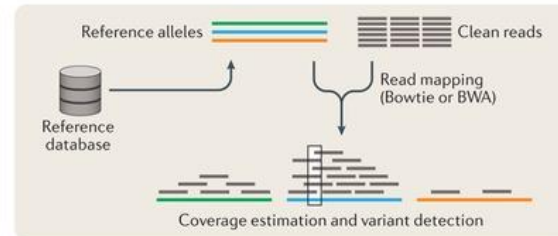
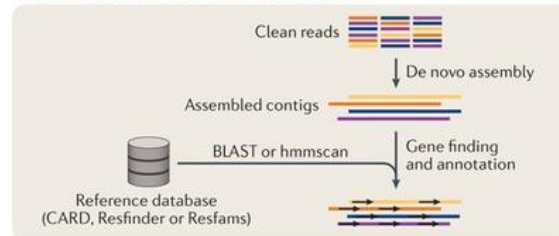
Culture-independent method (metagenomics)



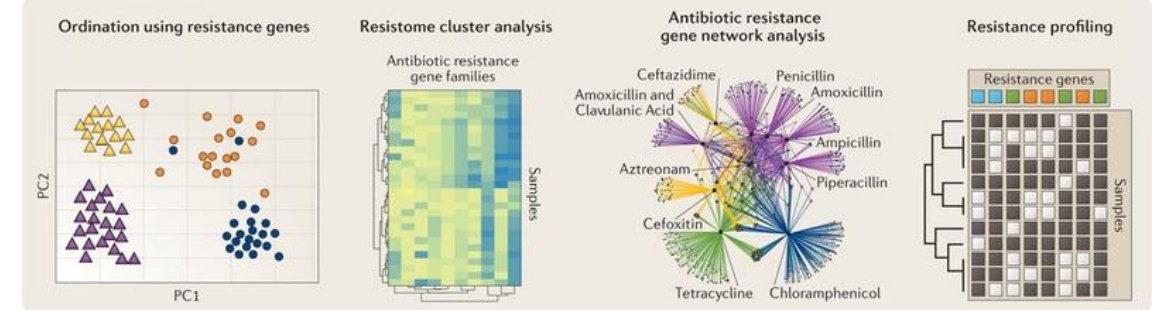
Culture-dependent method (isolate genomics)



b Assembly-based versus read-based approach



c Downstream analysis



Assembly-based approach:

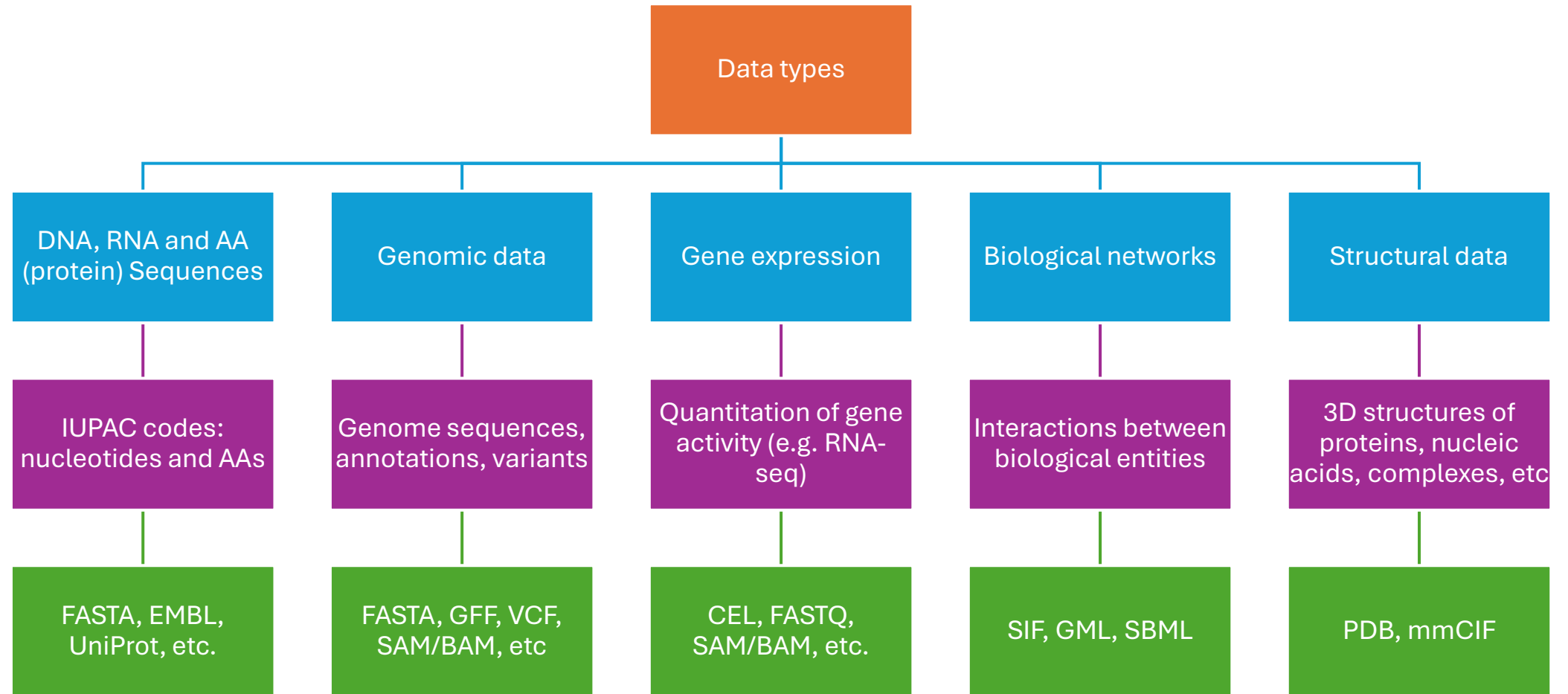
- Computationally expensive and time consuming, particularly in resistome profiling of large complex communities
- Identification of both known and novel resistance genes that share low similarity with reference database; however, requires high genome coverage
- Genomic context such as regulatory and mobile element sequences can be captured

Read-based approach:

- Fast and less computationally demanding, enabling resistome analysis of large data sets
- Identification of resistance genes is dependent on completeness of the reference database of organisms under analysis
- Nearby genes and genomic context are missing; may lead to false positives due to spurious mapping

Adapted from: Boolchandani, D'Souza and Dantas. Sequencing-based methods and resources to study AMR.
[10.1038/s41576-019-0108-4](https://doi.org/10.1038/s41576-019-0108-4)

(Some) Biological data types and formats



PS: This is NOT, by far, a comprehensive list of what is out there.

PS2: This is not a formal hierarchy, it is just to help you comprehend how it works.

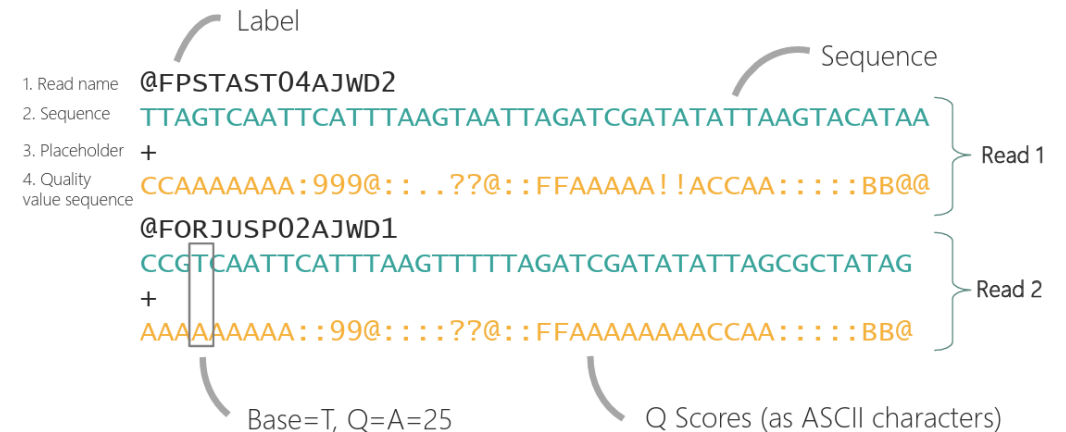
Sequence formats (FASTA and FASTQ)

- Both are ubiquitous text-based formats to represent sequences. They usually contain one or multiple sequences.
- Both begin with a label/ID for each sequence, followed by the sequence itself. FASTQ extends FASTA.
- Can be compressed.

```
>gi|1817694395|ref|NZ_JAAGMU010000151.1| Streptomyces sp. SID7958 contig-52000002, whole genome shotgun sequence
CCGGCTGGCGCGGCTGGCGCTGGCGGTGGGGCTGCGGCTGCTGGAGCTGGGGGTGGCGCTGGAGGCGCAC
GGCCAGAACCTGCTGGTGGTGTGTCGCCGTCCGGGAGCCGCGGCGGCTGGTCTACCGCGATCTGGCGG
ACATCCGGGTCTCCCCCGCGCGGCTGGCCCGGCACGGTATCCGGTTCCGGACCTGCCGGCG

>gi|1643051563|gb|SZWM01000399.1| Citrobacter sp. TBCS-14 contig3128, whole genome shotgun sequence
GCACAGTGAGATCAGCATTCGGTTGGATCTACTGGTCAATCAAAACCTGACGCTGGGTACTGAATGGAAC
CAGCAGCGCATGAAGGACATGCTGTCTAACTCGCAGACCTTTATGGGCGGTAATATTCCAGGCTACAGCA
GCACCGATCGCAGCCCATATTGAAAGCCGAGATCTTCTCTTTGTTGCCGAAACAACATG
```

FASTA is a common format for assembled genomes.
Extensions: “.fasta”, “.fna”, “.faa”, “.frn”



FASTQ is the *de facto* standard for data from NGS and 3rd generation sequencing technologies.
We will refer to them very often, also as “reads”. 10

ASCII table and corresponding Q-scores

Symbol	ASCII Code	Q-Score	Symbol	ASCII Code	Q-Score
!	33	0	6	54	21
"	34	1	7	55	22
#	35	2	8	56	23
\$	36	3	9	57	24
%	37	4	:	58	25
&	38	5	;	59	26
'	39	6	<	60	27
(	40	7	=	61	28
)	41	8	>	62	29
*	42	9	?	63	30
+	43	10	@	64	31
,	44	11	A	65	32
-	45	12	B	66	33
.	46	13	C	67	34
/	47	14	D	68	35
0	48	15	E	69	36
1	49	16	F	70	37
2	50	17	G	71	38
3	51	18	H	72	39
4	52	19	I	73	40
5	53	20			

Q scores represent the sequencer's estimate of the probability that the nucleotide at the position was called incorrectly.

$$Q = -10 \log_{10} p$$

p is the probability that the position was called incorrectly.

In this sense if:

Q=10 (ASCII: +), error probability is 1 in 10 (10%=0.1)

Q=20 (ASCII: 5), error probability is 1 in 100 (1%=0.01)

Q=30 (ASCII: ?), error probability is 1 in 1000 (0.1%=0.001).

Don't focus on memorizing symbols, but rather their meaning.

Alignment formats (SAM/BAM/CRAM)

- These represent how a sequence is aligned relative to a reference sequence (i.e. involves coordinates).

<pre>@HD VN:1.5 S0:coordinate @SQ SN:ref LN:45</pre>										
<pre>r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG * r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA * r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0; r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC * r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1; r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1</pre>										
Optional fields in the format of TAG:TYPE:VALUE QUAL: read quality; * meaning such information is not available SEQ: read sequence TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines. PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column. RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column. CIGAR: summary of alignment, e.g. insertion, deletion MAPQ: mapping quality POS: 1-based position RNAME: reference sequence name, e.g. chromosome/transcript id FLAG: indicates alignment information about the read, e.g. paired, aligned, etc. QNAME: query template name, aka. read ID										

Binary Alignment Map (BAM) is a binary representation of SAM, containing the same information in binary format for **improved performance**.

CRAM is a sequencing read file format that is highly space efficient by using reference-based compression of sequence data. Files are typically 30 to 60% **smaller** than their BAM equivalents

Variant information formats

- The Variant Call Format (VCF) is a format for storing variations (e.g., polymorphisms) between a reference genome and sequences aligned to it, based on SAM/BAM alignments

Binary Call Format (BCF) is a binary representation of VCF, containing the same information in binary format for **improved performance**.

Example

VCF header									
<pre>##fileformat=VCFv4.0 ##fileDate=20100707 ##source=VCFtools ##reference=NCBI36 ##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele"> ##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership"> ##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"> ##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)"> ##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)"> ##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth"> ##ALT=<ID=DEL,Description="Deletion"> ##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant"> ##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant"></pre>									
Body									
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1 SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13 0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0/1:100 2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1/0:77 1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3 0/0:20

Annotations:

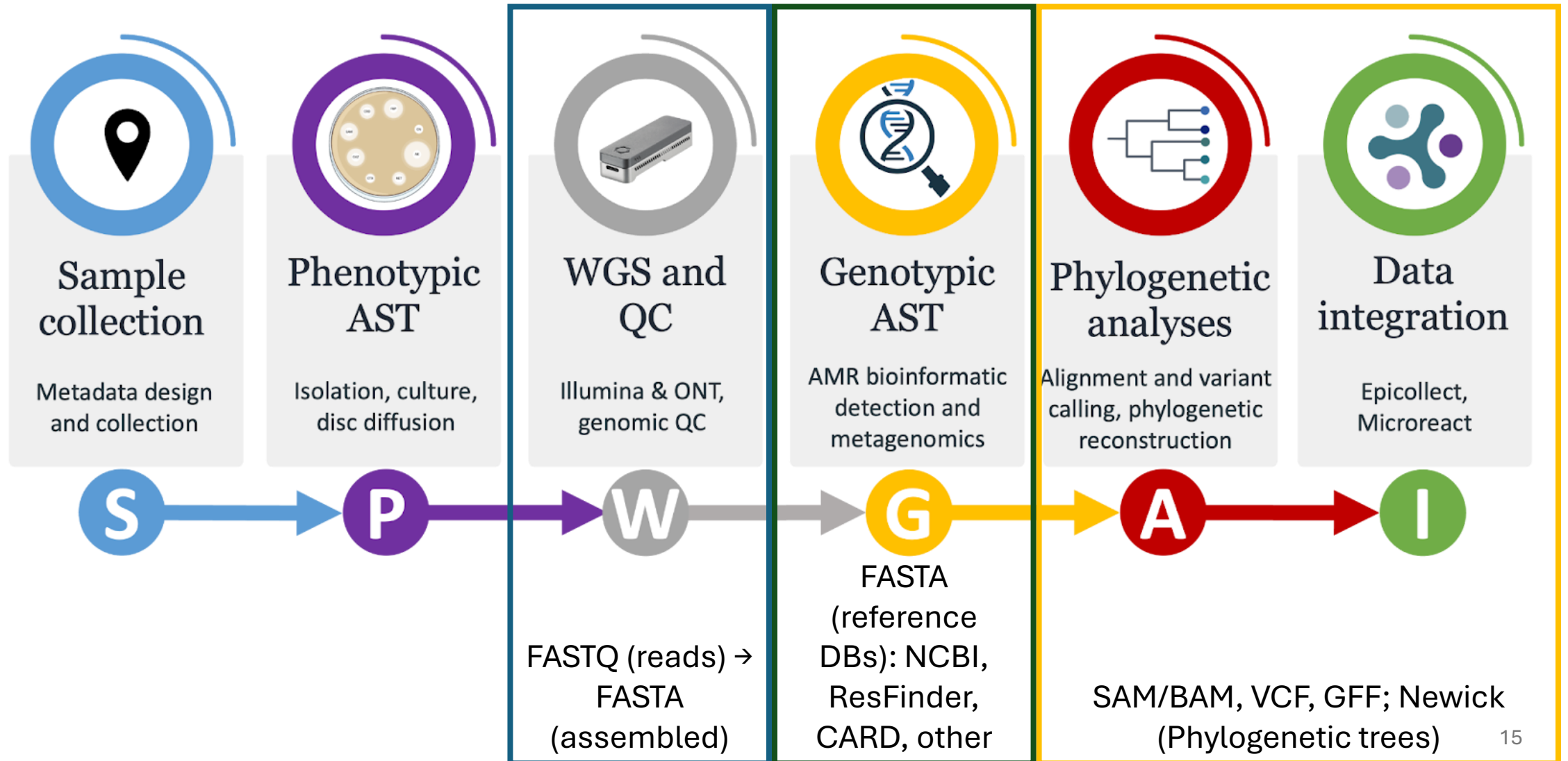
- Mandatory header lines:** ##fileformat=VCFv4.0
- Optional header lines (meta-data about the annotations in the VCF body):** ##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
- Reference alleles (GT=0):** A,AT
- Alternate alleles (GT>0 is an index to the ALT column):** T,CT
- Deletion:**
- SNP:** C → T
- Large SV:** SVTYPE=DEL;END=300
- Insertion:** T → CT
- Other event:** A → G
- Phased data (G and C above are on the same chromosome):** 1/1:12:3

Annotation formats

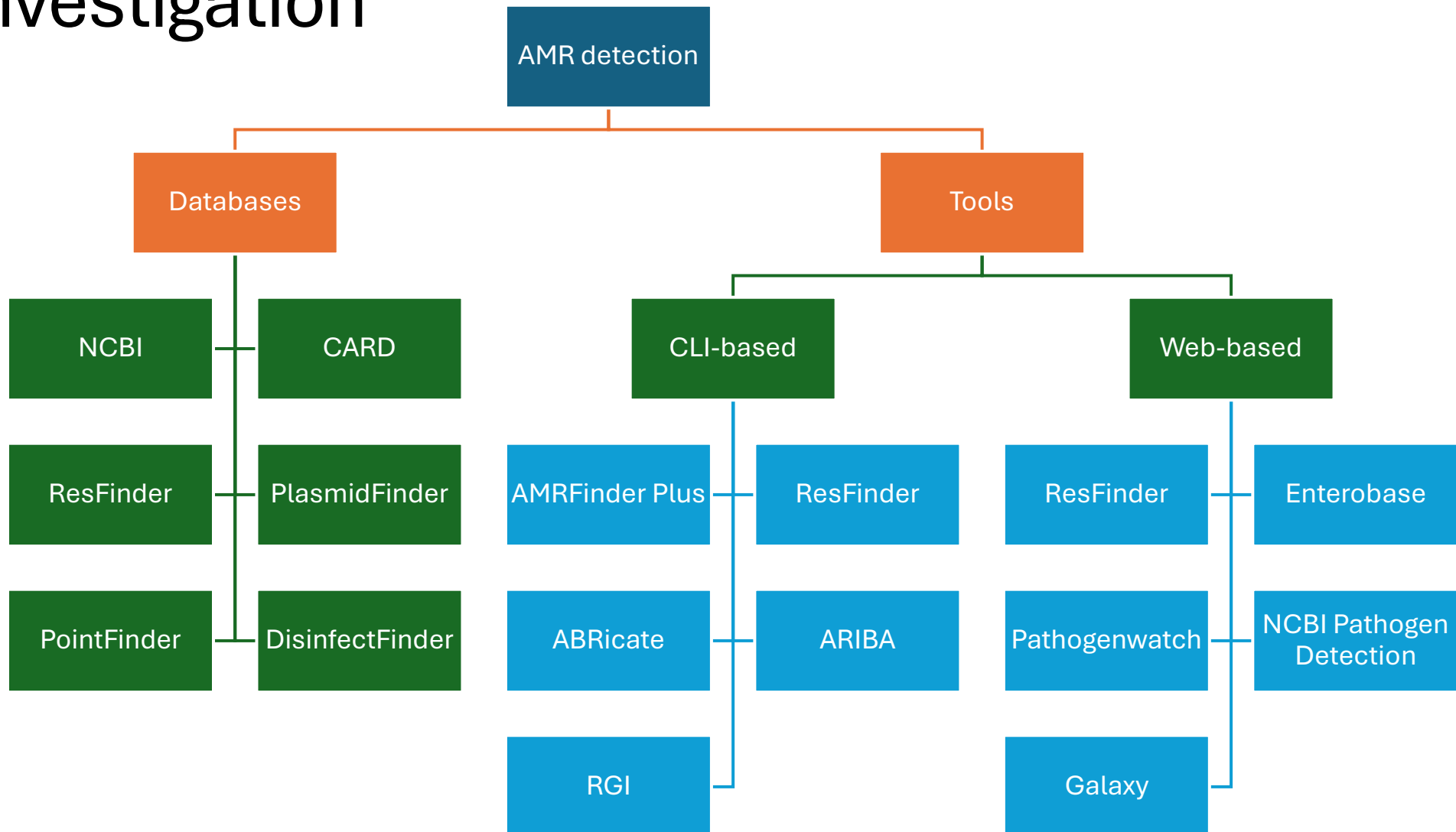
- Annotation: is the process of identifying and labelling/tagging the regions of the genome that correspond to functional elements.
- Feature: is the term used to refer to these functional elements that were identified through annotation and giving them the coordinates, names and more metadata.

```
##gff-version 3.2.1
##sequence-region ctg123 1 1497228
ctg123 . gene      1000  9000  .  +  .  ID=gene00001;Name=EDEN
ctg123 . TF_binding_site 1000  1012  .  +  .  ID=tfbs00001;Parent=gene00001
ctg123 . mRNA      1050  9000  .  +  .  ID=mRNA00001;Parent=gene00001;Name=EDEN.1
ctg123 . mRNA      1050  9000  .  +  .  ID=mRNA00002;Parent=gene00001;Name=EDEN.2
ctg123 . mRNA      1300  9000  .  +  .  ID=mRNA00003;Parent=gene00001;Name=EDEN.3
ctg123 . exon      1300  1500  .  +  .  ID=exon00001;Parent=mRNA00003
ctg123 . exon      1050  1500  .  +  .  ID=exon00002;Parent=mRNA00001,mRNA00002
ctg123 . exon      3000  3902  .  +  .  ID=exon00003;Parent=mRNA00001,mRNA00003
ctg123 . exon      5000  5500  .  +  .  ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon      7000  9000  .  +  .  ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS       1201  1500  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
```

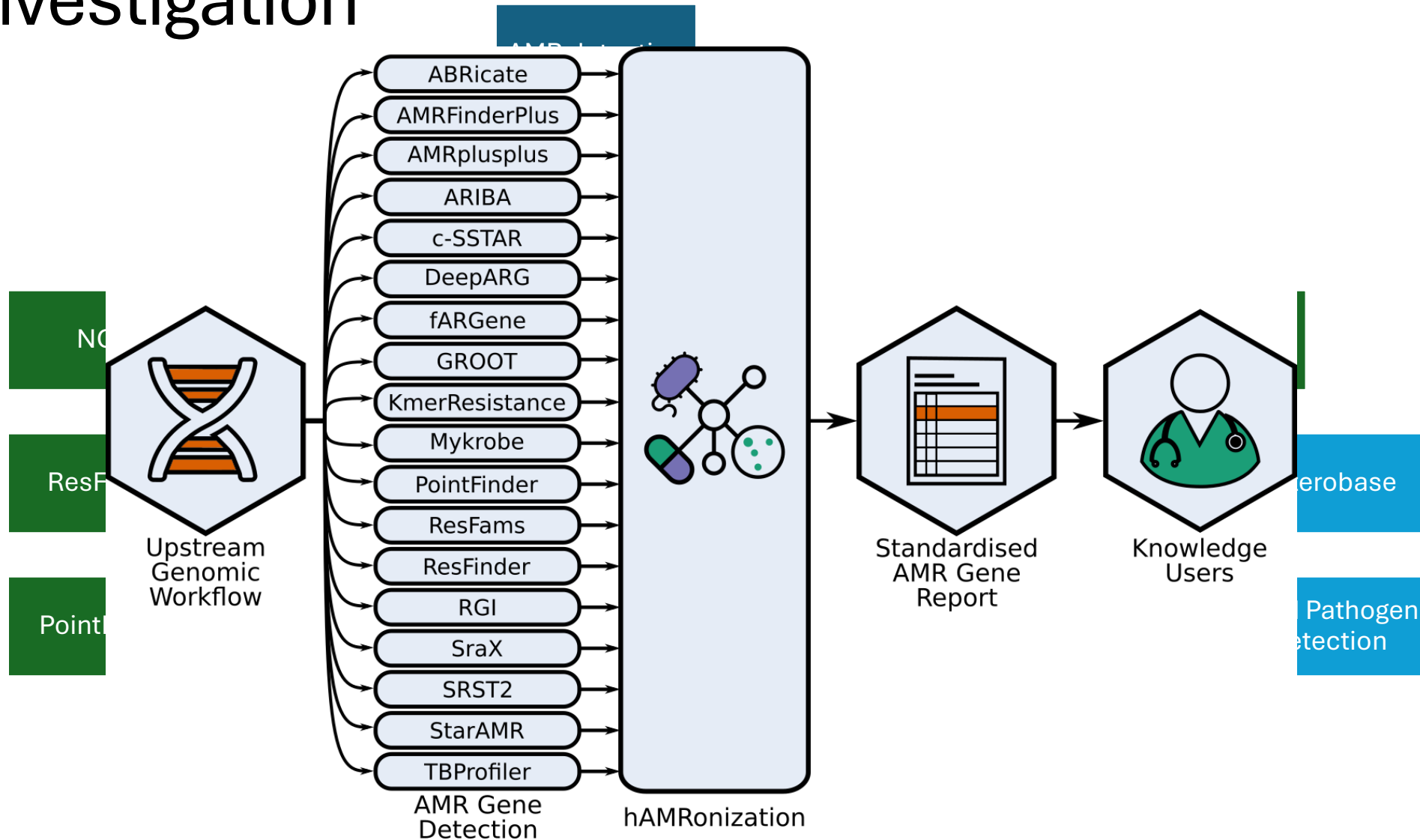
Workflow from sample to AMR report



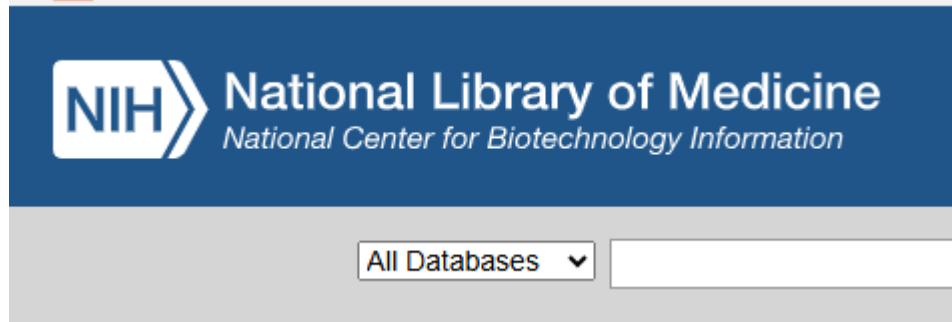
Computational Methods and DBs for AMR investigation



Computational Methods and DBs for AMR investigation



Sequence Databases (used often)



Genomes

Genome sequence assemblies, large-scale functional genomics data, and source biological samples

Assembly

Genome assembly information

BioCollections

Museum, herbaria, and other biorepository collections

BioProject

Biological projects providing data to NCBI

BioSample

Descriptions of biological source materials

Genome

Genome sequencing projects by organism

Nucleotide

DNA and RNA sequences

SRA

High-throughput sequence reads

Taxonomy

Taxonomic classification and nomenclature

BLAST

A tool to find regions of similarity between biological sequences

blastn

Search nucleotide sequence databases

blastp

Search protein sequence databases

blastx

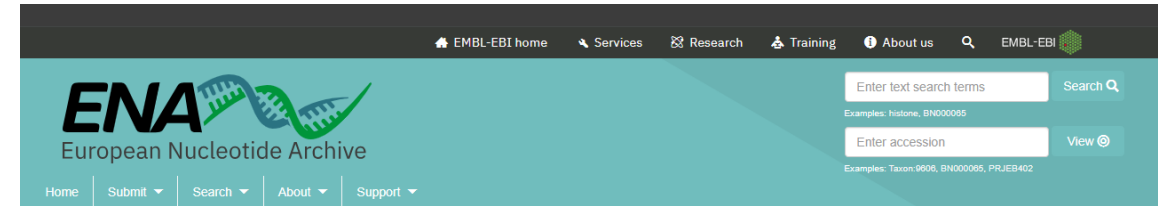
Search protein databases using a translated nucleotide query

tblastn

Search translated nucleotide databases using a protein query

Primer-BLAST

Find primers specific to your PCR template



European Nucleotide Archive

The European Nucleotide Archive (ENA) provides a comprehensive record of the world's nucleotide sequencing information, covering raw sequencing data, sequence assembly information and functional annotation. [More about ENA.](#)

Access to ENA data is provided through the browser, through search tools, through large scale file download and through the API.



ebi.ac.uk/ena/browser/home

Suggested readings

- Boolchandani, M., D'Souza, A.W. & Dantas, G. Sequencing-based methods and resources to study antimicrobial resistance. Nat Rev Genet 20, 356–370 (2019). Doi: [10.1038/s41576-019-0108-4](https://doi.org/10.1038/s41576-019-0108-4)
- Konopka, A.K., Crabbe, M.J. Introduction: Computational Biology in 14 Brief Paragraphs. In Konopka, A.K., Crabbe, M.J, editors. Compact Handbook of Computational Biology. CRC Press. 2004, 6 p.
- <https://fiveable.me/computational-biology/unit-1>
- <https://www.biosparked.com/course/niche-eszfp-zw9hf-ygfzr>
- https://rnnh.github.io/bioinfo-notebook/docs/file_formats.html
- <https://samformat.pages.dev/sam-format-flag>
- <https://support.nlm.nih.gov/kbArticle/?pn=KA-03574>

Acknowledgements and Partners



thank you!

Please contact wellcomeconnectingscience.org
for more information.

May 2021